

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CL)

April, 2013

CL 207 Concrete Technology

Date: 22.04.2013, Monday

Time: 01:30p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Which constituents present in mixing water change properties of concrete? [01]
(b) What are the advantages of using ready mixed concrete instead of site mixed concrete? [06]
- Q - 2 (a) Give the classification of aggregate based on shape. How texture of aggregates affect compressive strength in concrete? [07]
(b) Why admixtures are used in concrete? Explain mechanism of superplasticizers. [07]

OR

- Q - 2 (a) State use of Fineness Modulus. Find the Fineness Modulus of aggregate for the following result of sieve analysis. [07]

I.S Sieve	40	20	10	4.75	2.36	1.18	600	300	150
% Passing	100	70	50	40	20	2	0	0	0

- (b) Explain the role of accelerators and air entraining agents in concrete. [07]
- Q - 3 (a) How porosity affects the strength and durability of concrete? [05]
(b) Explain maturity of concrete. How it measured? What are their practical uses in concrete industries? [05]
(c) What do you understand by segregation and bleeding? How these affect the property of concrete? [04]

OR

- Q - 3 (a) Write short note on curing in term of its need, types, its effect on compressive strength and other properties of concrete. [05]
(b) Write short note on needle vibration in term of its types, spacing and numbers of insertion, duration and precautions while using it in various concrete. [05]
(c) Explain causes of microcracking. [04]

SECTION – II

- Q - 4 (a) Enlist advantages and disadvantages of concrete use. Explain grade of concrete. [05]
(b) Calculate Gel/Space ratio and theoretical strength of concrete, made with 600gm of cement with W/C ratio 0.5 at its 100% hydration. [02]
- Q - 5 (a) Draw flow chart showing manufacturing of cement by wet process. What are the major Bogue's compounds of cement? Discuss their role in hydration of cement. [07]
(b) Compare the properties of ordinary Portland cement and Portland pozzolana cement. [07]

OR

- Q - 5 (a) What is soundness of cement? Explain the testing procedure to determine the soundness of cement. Also define flash set. [07]
(b) Enlist the different types of cement. Discuss about the properties and applications for any two types of cement in concrete construction. [07]
- Q - 6 (a) Which properties of concrete are to be considered for the mix design? [04]
(b) Grading of fine and coarse aggregate available at a site is given below. Combine this aggregate in to suitable proportion to give the specified grading given along with. [10]

IS sieve	Percentage passing		Specified Grading
	Fine Aggregate	Coarse Aggregate	As per type 3 MSA – 40 mm
40	100	100	100
20	100	95	98
10	100	44	52
4.75	95	08	40
2.36	88	0	31
1.18	72	0	24
600 micron m	48	0	17
300 micron m	21	0	11
150 micron m	02	0	2

OR

- Q - 6 (a) Differentiate the approach of Nominal mix to Design mix. [04]
Q - 6 (b) Design a concrete mix as per IS:10262 - 2009, I.S. method for following data : [10]

Characteristic strength at 28 days = 30MPa

Maximum size of aggregate = 20 mm

Degree of workability = 100 mm slump

Quality control / Supervision = Very good

Type of exposure = severe

Method of placing = Manual

Type of Aggregate = Crushed Angular

Maximum Cement Content = 450 Kg/ m³

Chemical Admixture Type = Superplasticizer, 1% by mass of cementitious material

Specific gravity of Cement = 3.15

Coarse aggregate = 2.80

Fine aggregate = 2.60

Superplasticizer = 1.15

It is found that sand falls in grade zone III.